

worst-case scenarios, since his analysis period includes the Great Depression, a world war, and the stagnation of the 1970s.

If people can meet lifestyle goals using a safe withdrawal rate determined from history, they can be reasonably confident about their retirement, too. Also, in all but this worst-case scenario—so the argument goes—retirees will enjoy further upside as the portfolio grows when using a conservative withdrawal rate.

Regarding upside, Kitces and Bengen both reference—as a statement of confidence in safe withdrawal rates—the fact that in 96 percent of the US historical simulations, the value of assets remaining after thirty years was higher than the retirement date amount (although this is not adjusted for inflation). Ultimately, the idea is to retire when you can meet your spending goals from your portfolio using what you consider to be a safe withdrawal rate.

The Safety-First School of Thought

The safety-first school of thought was originally derived from academic models of how people allocate their resources over a lifetime to maximize lifetime satisfaction. Academics have studied these models since the 1920s to figure out how rational people make optimal decisions. In the retirement context, the question to be answered is how to get the most lifetime satisfaction from limited financial resources. It is the basic fundamental question of economics: how do you optimize in the face of scarcity? In more recent history, Nobel Prize winners such as Paul Samuelson, Robert Merton, Franco Modigliani, and William Sharpe have explored these models.

Safety-first comes from a more academic foundation, so it is often described with mathematical equations in academic journals. As a result, it has been slow to enter the public consciousness. The safety-first approach is probably best associated with Professor Zvi Bodie from Boston University, whose popular books such as *Worry-Free Investing* and *Risk Less* and Prosper have brought these ideas alive to the public. Michael Zwecher's